

1 Virtual displacements and work

1.1 Virtual displacements

$\delta W = 0 \Rightarrow P_1 \delta x_1 + P_2 \delta x_2 + \dots + P_n \delta x_n = 0$

1.2 Virtual work

$\delta W = \delta U$
 $P_1 \delta x_1 + P_2 \delta x_2 + \dots + P_n \delta x_n = F_1 \delta e_1 + F_2 \delta e_2 + \dots + F_m \delta e_m$

1.3 Steps

- 1. Set up a coordinate system with the origin at a **fixed point**
- 2. Assign displacement coordinates to each force
- 3. Write out $\delta W = 0$ (virtual displacements) or $\delta W = \delta U$ (virtual work)
- 4. Express all coordinate systems to a common variable (usually given) using $\delta y = \frac{dy}{dx} \delta x$
- 5. Drop the common infinitesimal variable (δx)

6 Solve for the applied force

2 Virtual complementary work

$\delta W^* = \delta U^*$
 $\Delta_1 \delta F_1 + \Delta_2 \delta F_2 + \dots + \Delta_n \delta F_n = e_1 \delta F_1 + e_2 \delta F_2 + \dots + e_m \delta F_m$

2.1 Steps

- 1. Write $\delta W^* = \delta U^*$
- 2. Drop unwanted deflection terms (Δ_n), and leave only the desired deflection term from the question
- 3. Replace e_m with $\frac{F_m}{k_m}$
- 4. Find the spring forces $F_1, F_2, \dots, F_m$ using moments or force balance

3 Unit load method

$\Delta = \delta U^*$

3.1 Equilibrium equations

$\sum F_x = 0, \sum F_y = 0, \sum M = 0$

3.2 Degree of indeterminacy

$n = N_{\text{unknowns}} - 3$

3.3 Steps

- 1. Write:

$$\Delta = \sum \int_0^{L_1} \frac{M_i m_i}{EI_i} dx + \sum \frac{F_i f_i}{k_i}$$

- How to decide where does a segment end?
 - Force changes
 - Beam changes (E, I or cross-section)
- 2. Solve for unknown reaction forces
- 3. Find internal moments of each segment $M_1, M_2, \dots, M_n$
 - Take a cut at the segment
 - Wherever the cut is made, draw the internal moment, like M_1
 - Take $\sum M = 0$ about the cut to find M_1
- 4. Perform virtual load (for linear displacement) or moment (for angular displacement) analysis
 - Remove all forces
 - For statically indeterminate structures, replace the unknown force or moment with a unit load or moment.
 - Apply unit load (1 N) or moment (1 Nm) at the point of interest
 - Find unknown reaction forces
 - Find internal virtual moment in each segment $m_1, m_2, \dots, m_n$
- 5. Sub $M_1, M_2, \dots, M_n, m_1, m_2, \dots, m_n$ into Δ and evaluate the integral

4 Impact loads

4.1 Gradual release

$\Delta_s = \frac{mg}{k}$

4.2 Instant release

$\Delta_{\text{dyn}} = \frac{2mg}{k}$

4.3 Dropped from height

$\Delta_{\text{dyn}} = \Delta_s \left(1 + \sqrt{1 + \frac{2h}{\Delta_s}} \right)$

5 Brittle fracture

5.1 Strain energy release rate

$G = \frac{\pi a \sigma^2}{E'} k$

- a is **half** the crack length
- $k = 1$ for a thin plate (plane stress)
- $k = (1 - \nu^2)$ for a thick block (plane strain)

5.2 Stress intensity factor

$K_I = Y \sigma \sqrt{\pi a}$

5.3 Variation of fracture toughness

5.3.1 Bending

$K_Q = \frac{F_Q}{BW^{3/2}} f_1 \left(\frac{W}{l} \right)$

5.3.2 Compact tension

$K_Q = \frac{F_Q}{BW^{3/2}} f_2 \left(\frac{W}{l} \right)$

5.3.3 Valid test criterion

$\frac{F_Q}{F_Q^{\text{max}}} < 1.1$

5.3.4 Valid K_{IC} criteria

$B, a, (W - a) \geq 2.5 \left(\frac{K_{IQ}}{\sigma_Y} \right)^2$
 $0.45 < \frac{a}{W} < 0.55$

6 Maths

6.1 Derivatives

Chain rule:

$\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx}$

Product rule:

$\frac{d}{dx} (u \cdot v) = \frac{du}{dx} \cdot v + u \cdot \frac{dv}{dx}$

Quotient rule:

$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$

Standard derivatives:

$\frac{d}{dx} (\sin x) = \cos x$

$\frac{d}{dx} (\cos x) = -\sin x$

$\frac{d}{dx} (\tan x) = \sec^2 x$

$\frac{d}{dx} (\arcsin x) = \frac{1}{\sqrt{1-x^2}}$

$\frac{d}{dx} (\arccos x) = -\frac{1}{\sqrt{1-x^2}}$

$\frac{d}{dx} (\arctan x) = \frac{1}{1+x^2}$

$\frac{d}{dx} (\sec x) = \sec x \tan x$

$\frac{d}{dx} (\csc x) = -\csc x \cot x$

$\frac{d}{dx} (\cot x) = -\csc^2 x$

6.2 Integrals

$\int \frac{1}{x} = \ln|x|$

$\int \sin x \, dx = -\cos x$

$\int \cos x \, dx = \sin x$

$\int \tan x \, dx = \ln|\sec x|$

$\int \frac{1}{x^2+a^2} \, dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right)$

$\int \frac{1}{\sqrt{a^2-x^2}} \, dx = \arcsin\left(\frac{x}{a}\right)$

$\int \frac{1}{x^2-a^2} \, dx = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right|$

$\int \frac{1}{a^2-x^2} \, dx = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right|$

$\int \frac{1}{\sqrt{x^2-a^2}} \, dx = \ln|\sqrt{x^2-a^2} + x|$

$\int \sec x \, dx = -\ln|\sec x + \tan x|$

$\int \csc x \, dx = -\ln|\csc x + \cot x|$

$\int \cot x \, dx = \ln|\sin x|$

Integration by parts:

$\int u \, dv = uv - \int v \, du$

Preference for u:

- 1. L: log or ln.
- 2. I: Inverse trigonometric functions.
- 3. A: Algebraic functions.
- 4. T: Trigonometric functions.
- 5. E: Exponential functions.

6.3 Trigonometric identities

Quotient identities:

$\tan \theta = \frac{\sin \theta}{\cos \theta}$

$\cot \theta = \frac{\cos \theta}{\sin \theta}$

Reciprocal identities:

$\sin \theta = \frac{1}{\csc \theta}$

$\csc \theta = \frac{1}{\sin \theta}$

$\cos \theta = \frac{1}{\sec \theta}$

$\sec \theta = \frac{1}{\cos \theta}$

$\tan \theta = \frac{1}{\cot \theta}$

$\cot \theta = \frac{1}{\tan \theta}$

Pythagorean identities:

$\sin^2 \theta + \cos^2 \theta = 1$

$\sec^2 \theta - \tan^2 \theta = 1$

$\csc^2 \theta - \cot^2 \theta = 1$

Even/odd identities:

$\sin(-\theta) = -\sin \theta$

$\cos(-\theta) = \cos \theta$

$\tan(-\theta) = -\tan \theta$

$\cot(-\theta) = -\cot \theta$

$\csc(-\theta) = -\csc \theta$

$\sec(-\theta) = \sec \theta$

Co-function identities:

$\sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta$

$\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$

$\tan\left(\frac{\pi}{2} - \theta\right) = \cot \theta$

$\cot\left(\frac{\pi}{2} - \theta\right) = \tan \theta$

$\csc\left(\frac{\pi}{2} - \theta\right) = \sec \theta$

$\sec\left(\frac{\pi}{2} - \theta\right) = \csc \theta$

$\frac{\pi}{2}$ radians = 90°

Sum/difference identities:

$\sin(\theta \pm \phi) = \sin \theta \cos \phi \pm \cos \theta \sin \phi$

$\cos(\theta \pm \phi) = \cos \theta \cos \phi \mp \sin \theta \sin \phi$

$\tan(\theta \pm \phi) = \frac{\tan \theta \pm \tan \phi}{1 \mp \tan \theta \tan \phi}$

Double angle identities:

$\sin(2\theta) = 2 \sin \theta \cos \theta$

$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$

$\cos(2\theta) = 2 \cos^2 \theta - 1$

$\cos(2\theta) = 1 - 2 \sin^2 \theta$

$\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$

Half angle identities:

$\sin^2 \theta = \frac{1 - \cos(2\theta)}{2}$

$\cos^2 \theta = \frac{1 + \cos(2\theta)}{2}$

$\tan^2 \theta = \frac{1 - \cos(2\theta)}{1 + \cos(2\theta)}$

Sum to product of 2 angles:

$\sin \theta + \sin \phi = 2 \sin\left(\frac{\theta + \phi}{2}\right) \cos\left(\frac{\theta - \phi}{2}\right)$

$\sin \theta - \sin \phi = 2 \cos\left(\frac{\theta + \phi}{2}\right) \sin\left(\frac{\theta - \phi}{2}\right)$

$\cos \theta + \cos \phi = 2 \cos\left(\frac{\theta + \phi}{2}\right) \cos\left(\frac{\theta - \phi}{2}\right)$

$\cos \theta - \cos \phi = -2 \sin\left(\frac{\theta + \phi}{2}\right) \sin\left(\frac{\theta - \phi}{2}\right)$

Product to sum of 2 angles:

$\sin \theta \sin \phi = \frac{\cos(\theta - \phi) - \cos(\theta + \phi)}{2}$

$\cos \theta \cos \phi = \frac{\cos(\theta - \phi) + \cos(\theta + \phi)}{2}$

$\sin \theta \cos \phi = \frac{\sin(\theta + \phi) + \sin(\theta - \phi)}{2}$

$\cos \theta \sin \phi = \frac{\sin(\theta + \phi) - \sin(\theta - \phi)}{2}$

Law of sines:

$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

Law of cosines:

$a^2 = b^2 + c^2 - 2bc \cos A$

Area of a triangle:

$A = \frac{1}{2} ab \sin C$